Waves

Mark scheme and worked solutions for 90 questions: 45 multiple choice, 45 structured.

1 · Part 1

Core · General properties of waves · [1]

Answer: B

Explanation

Waves transfer energy without transferring matter, so the water (and the leaf on it) only moves up and down as each ripple passes. A is wrong because the leaf is not carried along with the wave. C is wrong because the passing ripples do make the water surface move. D is wrong because there is no net movement of the water in any direction.

2 · Part 1

Core · General properties of waves · [1]

Answer: A

Explanation

In a longitudinal wave the vibrations are parallel to the direction the wave travels; sound waves and seismic P-waves (primary waves) are modelled as longitudinal. B is wrong because seismic S-waves are transverse. C is wrong because light and water waves are both transverse. D is wrong because water waves are transverse.

3 · Part 1

Core · General properties of waves · [1]

Answer: D

Worked solution

Moving the end up and down 4.0 times each second makes 4.0 waves each second, so f = 4.0 Hz. Wave speed v = fλ = 4.0 Hz × 0.75 m = 3.0 m / s. Check: in 1 s, 4.0 wavelengths of 0.75 m each leave the student's hand, which is 3.0 m of wave each second.

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4 · Part 1 Core General properties of waves [1]

Answer: B

Explanation

The amplitude is measured from the rest position to a crest, which is half the crest-to-trough height: 12 / 2 = 6 cm. A crest to the next trough is half a wavelength, so the wavelength is 2 × 40 = 80 cm. A uses half a wavelength as the wavelength, C and D use the full crest-to-trough height as the amplitude.

5 · Part 1 Core General properties of waves [1]

Answer: C

Explanation

Water waves travel more slowly in shallow water, so the speed decreases. The frequency is set by the source and does not change, so from $v = f\lambda$ the wavelength must decrease. A has the speed change the wrong way, B and D wrongly change the frequency.

6 · Part 1 Extended General properties of waves [1]

Answer: A

Explanation

Diffraction through a gap is greatest when the width of the gap is similar to the wavelength. In A the gap equals the wavelength. In B, C and D the gap is 5, 3 and 10 times the wavelength, so the waves spread out less.

7 · Part 1 Extended General properties of waves [1]

Answer: D

Explanation

Diffraction at an edge increases as the wavelength increases, so longer-wavelength waves bend further round the end of the wall. A is wrong because amplitude does not affect how much the waves diffract. B is wrong because shorter wavelengths diffract less. C is wrong because the length of the wall does not change how waves bend round its end.

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8 · Part 1 Core Reflection of light [1]

Answer: C

Worked solution

Draw (in your head) the normal at 90° to the mirror. The ray is 35° from the mirror, so it is 90° − 35° = 55° from the normal: angle of incidence = 55°. The law of reflection says angle of reflection = angle of incidence = 55°.

9 · Part 1 Core Reflection of light [1]

Answer: B

Explanation

A plane mirror forms an image that is virtual, the same size as the object, and the same distance behind the mirror as the object is in front of it: 1.5 m behind the mirror. A is wrong because the image cannot be formed on a screen, so it is not real. C gives the distance between the boy and his image, not the distance behind the mirror. D is wrong because the image appears behind the mirror, not on its surface.

10 · Part 1 Extended Reflection of light [1]

Answer: D

Worked solution

At first, angle of incidence = angle of reflection = 25°, so the reflected ray makes 25° + 25° = 50° with the incident ray. After the mirror is turned, both angles are 35°, so the reflected ray makes 35° + 35° = 70° with the incident ray. The incident ray stays in the same place, so the reflected ray must have turned through 70° − 50° = 20°. This is a general rule: turning a mirror through an angle turns the reflected ray through twice that angle (here 10° and 20°).

11 · Part 1 Core Refraction of light [1]

Answer: A

Explanation

Light slows down when it passes from air into glass, so it bends towards the normal and the angle of refraction is smaller than the angle of incidence. B describes light passing from glass into air. C only happens when the angle of incidence is 0°. D is wrong because total internal reflection can only happen when light travels from the optically denser material (glass) towards the less dense one (air).

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12 · Part 1 Core Refraction of light [1]

Answer: C

Explanation

The critical angle is the angle of incidence, in the optically denser material, at which the refracted ray travels along the boundary (angle of refraction 90°). A is wrong because total internal reflection cannot happen for light in air going into glass. B is wrong because at 0° incidence the angle of refraction is 0°. D is wrong because angles are measured from the normal, and all the light is reflected only above the critical angle.

13 · Part 1 Core Refraction of light [1]

Answer: D

Explanation

In a prism periscope, light inside the glass meets the long face at an angle of incidence (45°) greater than the critical angle, so it is totally internally reflected and turned through 90°. A and C are caused by refraction at the water surface. B is ordinary reflection from a mirror surface.

14 · Part 1 Extended Refraction of light [1]

Answer: B

Worked solution

Use n = sin i / sin r, where i is the angle in air. sin 45° = 0.707 and sin 28° = 0.469, so n = 0.707 / 0.469 = 1.51, which is 1.5 to two significant figures. Check: a refractive index from air into a solid must be greater than 1, because light slows down in the solid, so A and D cannot be right.

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15 · Part 1 Extended Refraction of light [1]

Answer: C

Worked solution

Step 1: find the critical angle. sin c = 1 / n = 1 / 1.60 = 0.625, so c = 38.7°, about 39°.

Step 2: compare. The angle of incidence, 42°, is bigger than 39°, and the light is going from glass into air, so it is totally internally reflected.

Step 3: the law of reflection still applies, so the angle of reflection = 42°.

Check another way: if the ray did leave the glass, sin r = 1.60 × sin 42° = 1.60 × 0.669 = 1.07. No angle has a sine greater than 1, so no refracted ray is possible.

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16 · Part 1 Core General properties of waves [5]

Mark scheme

(a) **[1]**

- longitudinal (wave) (1)

(b) **[2]**

- (the tape) vibrates / oscillates backwards and forwards (1)
- along the length of the spring / parallel to the direction the wave travels (1)

ALLOW it returns to / stays near its original position, it does not travel along the spring, for the first mark

REJECT moves along the spring with the wave

(c) **[1]**

- energy (1)

REJECT matter / coils / the spring

(d) **[1]**

- at right angles / perpendicular (to the direction the wave travels) (1)

Worked solution

(a) When the coils are pushed and pulled along the length of the spring, the vibrations are parallel to the direction the wave travels, so this is a longitudinal wave (like sound).

(b) Each coil, and the tape on it, just vibrates backwards and forwards along the spring about its rest position. It does not travel along with the wave.

(c) A wave transfers energy from one place to another without transferring matter. The coils end up where they started.

(d) Moving the end from side to side makes a transverse wave: the coils vibrate at right angles to the direction the wave travels.

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17 · Part 1 Core General properties of waves [6]

Mark scheme

(a) **[1]**

- the number of waves passing a point per second / the number of vibrations (oscillations) per second (1)

> **ALLOW** number of waves produced (by a source) per second

(b) **[2]**

- f = 12 / 60 (1)
- 0.20 Hz (1)

> **ALLOW** 0.2 Hz

(c) **[2]**

- v = fλ / 0.20 × 7.5 (1)
- 1.5 m / s (1)

> **ECF** from (b)

(d) **[1]**

- 6.0 m (1)

> **ALLOW** 6 m

> **ECF** (answer to (c)) / 0.25

Worked solution

(a) Frequency is the number of waves passing a point each second (measured in hertz).

(b) 12 waves pass in 60 s, so the number per second is 12 / 60 = 0.20 Hz.

(c) The distance between neighbouring crests is the wavelength, 7.5 m. v = fλ = 0.20 Hz × 7.5 m = 1.5 m / s. Check: one wave takes 60 / 12 = 5.0 s to pass, and 7.5 m / 5.0 s = 1.5 m / s.

(d) Rearrange v = fλ to λ = v / f = 1.5 / 0.25 = 6.0 m. A higher frequency at the same speed gives a shorter wavelength, which agrees with 6.0 m being less than 7.5 m.

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18 · Part 1 Core General properties of waves [6]

Mark scheme

(a) **[1]**

- S-waves / secondary waves (1)

(b) **[3]**

- 360 km = 360 000 m, and time = distance / speed used for either wave (1)
- P-waves 45 s and S-waves 80 s (1)
- 35 s (1)

ALLOW final answer of 35 s with no working for 3 marks

ECF for a wrong distance conversion used consistently in both times, for the subtraction mark

(c) **[2]**

- $\lambda = v / f$ / 4500 / 0.50 (1)
- 9000 m (1)

ALLOW 9.0×10^3 m

Worked solution

(a) In S-waves the vibrations are at right angles to the direction of travel, so they are modelled as transverse. P-waves are longitudinal.

(b) Convert the distance: 360 km = 360 000 m. Time for P-waves = distance / speed = 360 000 / 8000 = 45 s. Time for S-waves = 360 000 / 4500 = 80 s. The S-waves arrive 80 − 45 = 35 s after the P-waves. Check: in 45 s the S-waves travel 4500 × 45 = 202 500 m, leaving 157 500 m, which takes 157 500 / 4500 = 35 s.

(c) $\lambda = v / f$ = 4500 m / s ÷ 0.50 Hz = 9000 m.

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19 · Part 1 Core General properties of waves [6]

Mark scheme

(a) **[2]**

- place a (glass / plastic) plate / sheet on the bottom of part of the tank to make the water shallower (1)
- the waves change direction at the boundary / the wavefronts bend (and get closer together) in the shallow water (1)

> **ALLOW** the wavelength decreases / the wavefronts are closer together in the shallow water, for the second mark

(b) **[2]**

- f = v / λ / 24 / 1.2 (1)
- 20 Hz (1)

(c) **[2]**

- frequency stays the same / 20 Hz used (1)
- 16 cm / s (1)

> **ECF** from (b): (answer to (b)) × 0.80

> **ALLOW** 0.16 m / s

Worked solution

(a) Put a flat sheet of glass or plastic on the bottom of one part of the tank, so the water above it is shallower. When straight waves hit the edge of the sheet at an angle, they change direction (refract), and the wavefronts are closer together in the shallow water because the waves slow down.

(b) f = v / λ = 24 cm / s ÷ 1.2 cm = 20 Hz.

(c) The frequency is set by the vibrating bar, so it is still 20 Hz in the shallow water. v = fλ = 20 × 0.80 = 16 cm / s. Check: the wavelength has fallen in the ratio 0.80 / 1.2, and so has the speed: 24 × 0.80 / 1.2 = 16 cm / s. The waves are slower in shallow water, as expected.

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20 · Part 1 Core General properties of waves [5]

Mark scheme

(a) **[3]**

- a vibrating (straight) bar / dipper produces straight (plane) waves (1)
- two barriers placed in the tank with a (narrow) gap between them, in the path of the waves (1)
- the waves are seen (e.g. using a lamp above the tank and the pattern on a screen / paper below) to spread out / curve beyond the gap (1)

(b) **[1]**

- it stays the same / does not change (1)

(c) **[1]**

- the waves bend / spread round the edge (into the region behind the barrier) (1)

Worked solution

(a) A vibrating straight bar makes straight waves. Two barriers are put across the tank with a narrow gap between them. A lamp above the tank casts the wave pattern onto a screen or paper below. The straight waves that pass through the gap are seen to spread out into curved waves on the other side: this is diffraction.

(b) Diffraction does not change the speed or the frequency of the waves, so the wavelength stays the same (the wavefronts are the same distance apart on both sides of the gap).

(c) At a single edge the waves also diffract: they bend round the edge and spread into the region behind the barrier.

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21 · Part 1 Extended General properties of waves [7]

Mark scheme

(a) **[2]**

- λ = v / f / 342 / 380 (1)
- 0.90 m (1)

ALLOW 0.9 m

(b) **[3]**

- the wavelength of the sound is similar to / about the same as the width of the doorway (1)
- the wavelength of light is much smaller than the width of the doorway (1)
- diffraction (spreading) is greatest when the gap width is similar to the wavelength / diffraction is very small when the gap is much wider than the wavelength (1)

ECF from (a) for the comparison of the sound wavelength with the doorway

(c) **[2]**

- the low-frequency note (1)
- it has a longer wavelength, and longer wavelengths diffract more at an edge (1)

REJECT explanations based on loudness / amplitude

Worked solution

(a) λ = v / f = 342 m / s ÷ 380 Hz = 0.90 m.

(b) The sound wavelength (0.90 m) is about the same as the width of the doorway (0.90 m), so the sound diffracts strongly and spreads out into the classroom, reaching the student even though the student is not in line with the teacher. Light has a wavelength of about 5.0×10^{-7} m, which is millions of times smaller than the doorway, so it hardly diffracts at all and travels in straight lines through the gap. The student cannot see the teacher because no light from her travels in a straight line to the student's eyes.

(c) For the same speed of sound, v = fλ means a lower frequency gives a longer wavelength. Longer wavelengths diffract more at an edge, so the low note bends further round the door frame.

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22 · Part 1 Core Reflection of light [6]

Mark scheme

(a) **[1]**

- a line at right angles / perpendicular to the (mirror) surface at the point where the ray strikes it (1)

(b) **[2]**

- angle of incidence = angle of reflection / 64 / 2 (1)
- 32° (1)

(c) **[2]**

- any two of: virtual (1); same size (as the girl) (1); upright (1); laterally inverted (left and right swapped) (1)

REJECT real / magnified / diminished

(d) **[1]**

- 5.0 m (1)

ALLOW 5 m

Worked solution

(a) The normal is an imaginary line drawn at 90° to the mirror where the ray hits it. Angles of incidence and reflection are measured from it.

(b) The angle between the two rays is split equally by the normal, because angle of incidence = angle of reflection. So the angle of incidence = 64° / 2 = 32°.

(c) An image in a plane mirror is virtual (it cannot be formed on a screen), the same size as the object, upright and laterally inverted.

(d) The girl is 2.5 m in front of the mirror and the image is 2.5 m behind it, so they are 2.5 + 2.5 = 5.0 m apart.

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23 · Part 1 Extended Reflection of light [5]

Mark scheme

(a)(i) **[1]**

- 45° (1)

(a)(ii) **[1]**

- 90° (1)

> **ECF** 2 × answer to (a)(i)

(b) **[3]**

- distance walked = 1.2 × 2.5 = 3.0 m (1)
- distance from mirror = 4.0 − 3.0 = 1.0 m (1)
- distance between man and image = 2.0 m (1)

> **ECF** at each step from an earlier error

Worked solution

(a)(i) The normal is at 90° to the mirror surface. The mirror is at 45° to the horizontal, so the normal is also at 45° to the horizontal ray. The angle of incidence is 45°.

(a)(ii) The angle of reflection is also 45°, so the angle between the incident and reflected rays is 45° + 45° = 90°: the ray now travels vertically. This is how a periscope turns light through a right angle.

(b) Distance walked = speed × time = 1.2 m / s × 2.5 s = 3.0 m. He is now 4.0 − 3.0 = 1.0 m from the mirror. His image is the same distance behind the mirror, 1.0 m, so the distance between the man and his image is 1.0 + 1.0 = 2.0 m. Check: at the start the man and image were 8.0 m apart; each metre he walks closes the gap by 2 m, so 8.0 − 2 × 3.0 = 2.0 m.

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24 · Part 1 Core Refraction of light [6]

Mark scheme

(a) **[4]**

- place the block on the paper and draw round its outline (1)
- shine a ray from the ray box into one face of the block at an angle (not along the normal) (1)
- mark the path of the incident ray and the emerging ray (e.g. with crosses / dots), remove the block and join the points, including across the block (1)
- draw the normal where the ray enters and measure the angles between the normal and the incident ray and between the normal and the refracted ray with the protractor (1)

(b) **[1]**

- parallel (to the incident ray) / in the same direction (but shifted sideways) (1)

(c) **[1]**

- the ray meets the curved surface along the normal / the angle of incidence is 0° (1)

ALLOW a radius is at right angles to the curved surface

Worked solution

(a) Put the glass block on the paper and draw round it. Aim a narrow ray from the ray box at one of the long faces at an angle. Mark the incoming ray with two crosses and the ray coming out of the opposite face with two crosses. Take the block away, draw lines through the crosses and join the point where the ray entered to the point where it left: this line is the path inside the glass. At the point of entry, draw the normal at 90° to the face. Use the protractor to measure the angle of incidence (between the incident ray and the normal) and the angle of refraction (between the ray in the glass and the normal).

(b) The ray bends towards the normal on entering and away from the normal by the same amount on leaving, because the two faces are parallel. So the ray that leaves is parallel to the ray that entered, just shifted sideways.

(c) A radius of a circle meets the curved edge at 90°, so a ray along a radius is along the normal. Its angle of incidence is 0°, so it does not bend.

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25 · Part 1 Core Refraction of light [5]

Mark scheme

(a) **[1]**

- the angle between the refracted ray and the normal (1)

(b) **[1]**

- straight on / along the normal / it does not change direction (1)

(c) **[2]**

- greater than 25° (1)
- light speeds up / travels faster in air than in glass, so it bends away from the normal (1)

> **ALLOW** it goes from an optically denser material into a less dense one, so it bends away from the normal

(d) **[1]**

- it decreases / the light slows down (1)

Worked solution

(a) The angle of refraction is measured between the refracted ray and the normal (not the surface).

(b) A ray that meets a boundary along the normal (angle of incidence 0°) changes speed but not direction, so it carries straight on downwards.

(c) Light travels faster in air than in glass. When it passes from glass into air at an angle, it bends away from the normal, so the angle of refraction is greater than the angle of incidence: more than 25°.

(d) Light travels more slowly in glass than in air, so its speed decreases as it enters the glass. This is why it bends towards the normal on entering.

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26 · Part 1 Core Refraction of light [6]

Mark scheme

(a) **[1]**

- the angle of incidence (in the glass / denser material) at which the angle of refraction is 90° (1)

(b) **[3]**

- at small angles, the ray passes out into the air and bends away from the normal, with a weak reflected ray (1)
- at the critical angle, the refracted ray travels along the flat face / the angle of refraction is 90° (1)
- above the critical angle, there is no ray in the air: all the light is (totally internally) reflected back into the glass (1)

ALLOW at 0°, the ray passes straight out without bending, as part of the first mark

(c) **[2]**

- the light travels from an optically denser material to a less dense one / from glass to air (1)
- the angle of incidence is greater than the critical angle (1)

Worked solution

(a) The critical angle is the angle of incidence inside the glass at which the refracted ray just runs along the boundary, at 90° to the normal.

(b) At 0° the ray passes straight out. As the angle of incidence increases, the ray leaving the flat face bends away from the normal more and more, and a faint reflected ray is also seen inside the glass. When the angle of incidence reaches the critical angle, the refracted ray travels along the flat face. Above the critical angle no light leaves the glass: all of it is reflected back inside. This is total internal reflection.

(c) Total internal reflection happens only when light is travelling from the optically denser material towards the less dense one (glass to air), and the angle of incidence is bigger than the critical angle.

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27 · Part 1 Extended Refraction of light [6]

Mark scheme

(a) **[1]**

- the ratio of the speeds of a wave in two different regions (materials) (1)

> **ALLOW** speed of the wave in the first region / speed in the second region, or speed of light in air (vacuum) / speed of light in the material

(b) **[2]**

- n = 6.0 / 4.0 (1)
- 1.5 (1)

> **REJECT** 0.67

> **IGNORE** a unit

(c) **[3]**

- sin r = sin i / n / sin r = sin 40° / 1.5 (1)
- sin r = 0.43 (1)
- r = 25° (1)

> **ECF** from (b)

Worked solution

(a) The refractive index is the ratio of the speeds of a wave in two different regions: speed in the first region divided by speed in the second.

(b) n = speed in deeper water / speed in shallower water = 6.0 / 4.0 = 1.5. It is greater than 1 because the waves slow down, just as light slows down when it enters glass.

(c) Rearrange n = sin i / sin r to sin r = sin i / n = sin 40° / 1.5 = 0.643 / 1.5 = 0.429. So r = 25°. Check: sin 40° / sin 25° = 0.643 / 0.423 = 1.5. The angle of refraction is smaller than the angle of incidence, so the waves bend towards the normal as they slow down, which makes sense.

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28 · Part 1 Extended Refraction of light [8]

Mark scheme

(a) **[2]**

- light is (totally internally) reflected at the sides of the rod (1)
- repeatedly / many times, because the angle of incidence at the sides is greater than the critical angle (1)

(b) **[2]**

- sin c = 1 / n / sin c = 1 / 1.48 / sin c = 0.676 (1)
- 42.5° (1)

ALLOW 42° or 43°

(c) **[2]**

- the ray refracts / passes out of the rod into the air (light escapes / is lost), with only partial reflection (1)
- because 40° is less than the critical angle (1)

ECF from (b) if the comparison is consistent with the critical angle calculated

(d) **[2]**

- any two of: they can carry a large amount of information / many signals at the same time / data at a high rate (1); signals can travel long distances with little loss of energy / little weakening (1); the signals are not affected by electrical interference (1); signals are hard to tap / more secure (1); the fibres are thin and light (1)

Worked solution

(a) Light inside the rod hits the sides at a large angle of incidence, greater than the critical angle, so it is totally internally reflected. This happens again and again, so the light zig-zags along the rod and comes out at the far end, even if the rod is gently curved.

(b) sin c = 1 / n = 1 / 1.48 = 0.676, so c = 42.5°.

(c) The angle of incidence, 40°, is smaller than the critical angle (42.5°), so total internal reflection does not happen. Most of the light refracts out through the side of the rod into the air, and only a small amount is reflected. This light is lost from the rod.

(d) Optical fibres carry signals as pulses of light. One fibre can carry a huge amount of information at the same time, the signals lose very little energy over long distances, and they are not affected by electrical interference.

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29 · Part 1 Extended Refraction of light [5]

Mark scheme

(a) **[2]**

- n = sin 40° / sin 25° / 0.643 / 0.423 (1)
- 1.5 (1)

ALLOW 1.52

(b) **[2]**

- sin r = sin 50° / n / sin r = 0.766 / 1.5 (= 0.51) (1)
- 30° to 31° (1)

ECF from (a)

(c) **[1]**

- 0° (1)

Worked solution

(a) n = sin i / sin r = sin 40° / sin 25° = 0.643 / 0.423 = 1.52, which is 1.5 to two significant figures. The other two rows give almost the same value (sin 20° / sin 13° = 1.52 and sin 60° / sin 35° = 1.51), which shows n is a constant for the acrylic.

(b) Rearrange n = sin i / sin r to sin r = sin i / n = sin 50° / 1.5 = 0.766 / 1.5 = 0.511, so r = 31° (using n = 1.52 gives 30°). Check: the answer lies between the results for 40° (25°) and 60° (35°), as it should.

(c) A ray along the normal has an angle of incidence of 0°, so it does not bend: the angle of refraction is 0°.

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30 · Part 1 Core Refraction of light [6]

Mark scheme

(a) **[1]**

- it enters along the normal / the angle of incidence is 0° (1)

(b) **[2]**

- the light is travelling from glass to air / from a denser to a less dense material (1)
- the angle of incidence (45°) is greater than the critical angle (42°) (1)

(c) **[1]**

- 90° (1)

(d) **[2]**

- 45° is less than the critical angle (48°), so the ray is not totally internally reflected / it refracts out of the longest face into the air (1)
- it bends away from the normal / the angle of refraction is greater than 45° (1)

ALLOW a weak (partially) reflected ray is also produced, for the second mark

Worked solution

(a) The ray meets the first face at right angles, so it travels along the normal. Its angle of incidence is 0°, so it does not bend.

(b) Inside the prism the light is going from glass towards air. Its angle of incidence at the long face (45°) is bigger than the critical angle (42°), so no light can refract out and all of it is reflected back into the glass.

(c) The angle of incidence and the angle of reflection are both 45°, so the ray is turned through 45° + 45° = 90°. It then leaves through the other short face at right angles.

(d) For the plastic, 45° is smaller than the critical angle of 48°, so total internal reflection does not happen. Most of the light refracts out of the long face into the air, bending away from the normal (angle of refraction more than 45°), and only a weak ray is reflected. The plastic prism would not work well in binoculars.

1 · Part 2 Core Thin lenses [1]

Answer: B

Explanation

A diverging lens spreads a parallel beam out so that the rays appear to come from the principal focus on the same side as the incoming light. A describes a converging lens. C is wrong because the rays do not stay parallel. D puts the apparent source on the wrong side: the rays only appear to come from a point if they are traced backwards towards the incoming light.

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2 · Part 2 Core Thin lenses [1]

Answer: D

Explanation

The principal focus is the point where rays travelling parallel to the principal axis meet after passing through a converging lens. A describes the optical centre of the lens. B is wrong because the image position changes with the object position. C describes the point at twice the focal length, not the principal focus.

3 · Part 2 Core Thin lenses [1]

Answer: A

Explanation

The object is 30 cm away, which is more than twice the focal length (20 cm). The image is real (it forms on a screen), inverted and diminished. B is the result for an object between one and two focal lengths. C is the result for an object closer than the principal focus, and that image could not form on a screen. D is wrong because a real image formed by a single converging lens is always inverted.

4 · Part 2 Core Thin lenses [1]

Answer: C

Explanation

A virtual image is where diverging rays appear to come from when they are extended backwards; no light actually reaches that point, so it cannot be projected onto a screen. A and B describe a real image. D is wrong because an object more than two focal lengths from a converging lens gives a real image.

5 · Part 2 Extended Thin lenses [1]

Answer: A

Explanation

A short-sighted eye sees near objects clearly but focuses light from distant objects in front of the retina. A diverging lens spreads the rays out a little before they enter the eye, so they focus on the retina. B uses the wrong lens, which would make the problem worse. C and D describe light from near objects focusing behind the retina, which is long-sightedness, not short-sightedness.

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6 · Part 2 Core Dispersion of light [1]

Answer: B

Explanation

In dispersion each colour is refracted by a different amount. Violet is refracted the most and red the least. Green and yellow are refracted by amounts between those of red and violet.

7 · Part 2 Core Dispersion of light [1]

Answer: D

Explanation

In order of increasing wavelength the spectrum runs violet, indigo, blue, green, yellow, orange, red, so blue, green, yellow, orange is correct. A is in order of decreasing wavelength. In B violet should come before indigo. In C blue has a shorter wavelength than green, so it cannot come after it.

8 · Part 2 Extended Dispersion of light [1]

Answer: C

Explanation

Monochromatic light is visible light of a single frequency. A describes white light. B is wrong because passing through a prism does not make light a single frequency (white light is split into many). D is wrong because monochromatic light is still refracted by glass.

9 · Part 2 Core Electromagnetic spectrum [1]

Answer: A

Explanation

In order of increasing wavelength: gamma rays, X-rays, ultraviolet, visible light, infrared, microwaves, radio waves. Infrared lies between visible light and microwaves. Ultraviolet and X-rays have shorter wavelengths than visible light. Radio waves have longer wavelengths than microwaves.

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10 · Part 2 Core Electromagnetic spectrum [1]

Answer: D

Explanation

All electromagnetic waves travel at the same high speed in a vacuum, whatever their frequency or wavelength. A, B and C all wrongly link speed in a vacuum to frequency, wavelength or penetrating ability.

11 · Part 2 Core Electromagnetic spectrum [1]

Answer: B

Explanation

Thermal imaging cameras detect infrared radiation. Fake bank notes are detected with ultraviolet, not X-rays (A). Satellite television uses microwaves, not gamma rays (C). Mobile phones use microwaves, not ultraviolet (D).

12 · Part 2 Core Electromagnetic spectrum [1]

Answer: C

Explanation

Excessive exposure to gamma rays (and X-rays) can cause mutation or damage to cells in the body. Excessive infrared causes skin burns (A) and excessive microwaves cause internal heating of body cells (B); neither is the effect described. Visible light (D) is not listed as causing cell mutation.

13 · Part 2 Core Electromagnetic spectrum [1]

Answer: D

Explanation

Direct broadcast satellite television uses geostationary satellites, and communication with artificial satellites is mainly by microwaves. Low orbit satellites (A, B) are used by some satellite phones, not usually for direct broadcast television. Radio waves (A, C) are used for radio and land-based television transmissions, not mainly for satellite communication.

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14 · Part 2 Extended Electromagnetic spectrum [1]

Answer: A

Worked solution

Remember the two reasons together: "penetrate some walls" and "short aerial". Any option that says microwaves are faster than other electromagnetic waves can be ruled out straight away.

15 · Part 2 Extended Electromagnetic spectrum [1]

Answer: C

Worked solution

f = 120 MHz = 1.2×10^8 Hz. The speed of radio waves in air is approximately 3.0×10^8 m / s. Rearranging $v = f\lambda$ gives $\lambda = v / f = (3.0 \times 10^8) / (1.2 \times 10^8) = 2.5$ m. Check: 2.5 m × 1.2×10^8 Hz = 3.0×10^8 m / s.

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16 · Part 2 Core Thin lenses [5]

Mark scheme

(a) **[2]**

- the rays are refracted / bent towards the principal axis (1)
- the rays converge / meet at one point (on the principal axis) (1)

ALLOW "the rays come together at the principal focus" for both marks

REJECT "the rays are reflected"

(b) **[1]**

- principal focus (1)

ALLOW focal point / focus

(c) **[1]**

- 15 cm (1)

REJECT answer without a unit

(d) **[1]**

- the rays spread out / diverge (as if coming from the principal focus on the side the light enters) (1)

IGNORE reference to the rays being bent away from the principal axis

Worked solution

A converging lens refracts each ray of a parallel beam towards the principal axis, so all the rays meet at one point. That point is the principal focus (focal point). The distance from the centre of the lens to the principal focus is the focal length, so here the focal length is 15 cm. A diverging lens does the opposite: it spreads a parallel beam out, so the rays leaving it look as if they came from a point (its principal focus) on the same side as the incoming light.

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17 · Part 2 Core Thin lenses [6]

Mark scheme

(a) **[1]**

- the distance between the centre of the lens and the principal focus (1)

ALLOW focal point / focus for principal focus

REJECT "distance between the lens and the image" / "distance between the object and the lens"

(b) **[3]**

- real (1)
- inverted (1)
- enlarged (1)

ALLOW upside down for inverted; magnified for enlarged

- if more than three terms are given, each contradictory term (e.g. upright, virtual, diminished) cancels one correct mark

(c) **[1]**

- (the image is) the same size (as the filament) (1)

(d) **[1]**

- rays of light actually pass through / meet at the image (1)

ALLOW the image is formed on a screen

Worked solution

The focal length is the distance from the centre of the lens to the principal focus. Here the focal length is 8.0 cm, so twice the focal length is 16 cm.

In (b) the filament is 12 cm away, which is between one focal length (8.0 cm) and two focal lengths (16 cm). For an object in this range a converging lens forms a real, inverted, enlarged image on the far side of the lens.

In (c) the filament is exactly two focal lengths away (16 cm), so the image is real, inverted and the same size as the object.

Both images are real because the light rays really meet at the image, which is why they can be caught on a screen.

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18 · Part 2 Extended Thin lenses [6]

Mark scheme

(a) **[1]**

- closer to the lens than the principal focus / less than 5.0 cm from the lens (1)

ALLOW between the lens and the principal focus

(b) **[3]**

- virtual (1)
- upright (1)
- enlarged (1)

ALLOW the right way up for upright; magnified for enlarged

- if more than three terms are given, each contradictory term (e.g. real, inverted, diminished) cancels one correct mark

(c) **[2]**

- the rays leaving the lens diverge / do not actually meet (1)
- the image is only where the rays appear to come from when extended backwards / no light actually reaches the image position (1)

ALLOW "the image is virtual" for one mark only if no further explanation is given

Worked solution

A converging lens works as a magnifying glass when the object is placed closer to the lens than the principal focus, so here less than 5.0 cm away. The rays from the gemstone still spread out after passing through the lens, so they never meet. The eye traces them backwards and sees an image that is virtual, upright and enlarged, on the same side of the lens as the gemstone. Because no light actually arrives at the place where the image appears, a screen placed there would show nothing.

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19 · Part 2 Extended Thin lenses [6]

Mark scheme

(a) **[1]**

- long-sightedness (1)

ALLOW long sight / hypermetropia

(b) **[1]**

- behind the retina (1)

(c) **[1]**

- converging (lens) (1)

ALLOW convex

(d) **[2]**

- the lens refracts the rays (from the book) so that they converge / bend inwards before entering the eye (1)
- (so the eye can then) focus the light on the retina (1)

REJECT "the lens makes the image bigger" for either mark

(e) **[1]**

- her lens is diverging / spreads light out, but his converges light (1)

ALLOW her lens is concave and his is convex

Worked solution

The man sees distant objects clearly but not near ones, so he is long-sighted. In his eye, light from a nearby book is brought to a focus behind the retina, so the image on the retina is blurred. A converging lens starts to bend the rays inwards before they reach the eye. The eye then has less bending to do, so the light is focused on the retina and the words look sharp. His short-sighted daughter has the opposite problem (distant light focuses in front of her retina), so her spectacles have diverging lenses, which spread the light out a little before it enters the eye.

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20 · Part 2 Core Dispersion of light [6]

Mark scheme

(a) **[1]**

- dispersion (1)

REJECT refraction / diffraction on their own

(b) **[2]**

- white light is a mixture of colours / frequencies / wavelengths (1)
- each colour is refracted by a different amount (by the glass) (1)

ALLOW each colour changes direction by a different angle

(c) **[1]**

- red (1)

(d) **[2]**

- all seven colours named: red, orange, yellow, green, blue, indigo, violet (1)
- in the order red, orange, yellow, green, blue, indigo, violet (1)

ALLOW one mark only if the seven colours are given in the reverse order

Worked solution

White light is a mixture of all the colours of the visible spectrum. When it enters and leaves the glass prism it is refracted, but each colour is refracted by a slightly different amount, so the colours spread out into a spectrum. This is called dispersion. Red is refracted the least and violet the most. Red light has the lowest frequency and violet the highest, so in order of increasing frequency the colours are red, orange, yellow, green, blue, indigo, violet.

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21 · Part 2 Extended Dispersion of light [5]

Mark scheme

(a) **[1]**

- (visible) light of a single frequency (1)

> **ALLOW** light of a single wavelength

> **REJECT** "light of a single colour" on its own

(b) **[2]**

- a single (green) spot / patch, not a spectrum / band of colours (1)
- the light has only one frequency so it is all refracted by the same amount / no dispersion occurs (1)

> **ALLOW** the spot is in a different position from where it would be without the prism

- the explanation mark can only be scored if the prediction is a single colour

(c)(i) **[1]**

- violet (1)

(c)(ii) **[1]**

- red (1)

Worked solution

Monochromatic light is light of a single frequency. White light disperses in a prism because it is a mixture of frequencies and each frequency is refracted by a different amount. The laser light has only one frequency, so all of it is refracted by the same amount. It is still bent by the prism, but it does not spread out: the screen shows a single green spot, not a spectrum.

In a prism, violet light is refracted more than red light, so the violet beam changes direction by the greater amount. Red is at the long-wavelength end of the visible spectrum, so the red beam has the longer wavelength.

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22 · Part 2 Core Electromagnetic spectrum [5]

Mark scheme

(a) **[2]**

- P: infrared (1)
- Q: X-rays (1)

(b) **[1]**

- radio waves (1)

(c) **[1]**

- (they travel at) the same speed (1)

(d) **[1]**

- visible light (1)

Worked solution

In order of increasing frequency the regions are radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, gamma rays, so P is infrared and Q is X-rays. Wavelength gets shorter as frequency increases, so radio waves, at the low-frequency end, have the longest wavelength. All electromagnetic waves travel at the same high speed in a vacuum, so gamma rays and radio waves have the same speed. The human eye detects only visible light.

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23 · Part 2 Core Electromagnetic spectrum [6]

Mark scheme

(a)(i) **[1]**

- X-rays (1)

(a)(ii) **[1]**

- gamma rays (1)

(a)(iii) **[1]**

- infrared (1)

(a)(iv) **[1]**

- ultraviolet (1)

(b) **[2]**

- X-rays can cause mutation of / damage to cells in the body (1)
- the operator would be exposed every time a patient is scanned, so the total / repeated exposure would be excessive (the screen absorbs the X-rays) (1)

ALLOW "can cause cancer" for the first mark

REJECT "X-rays cause burns" / "X-rays heat the body"

Worked solution

X-rays are used for medical scanning, such as images of broken bones. Gamma rays are used to sterilise medical equipment, and they can do this even after the equipment has been sealed in its packet. Thermal imaging cameras detect infrared. Ultraviolet is used to detect fake bank notes.

A patient has only a few scans, but the operator works with the machine all day. Excessive exposure to X-rays can cause mutation or damage to cells in the body, so the operator stands behind a screen to avoid a large total dose built up over many scans.

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24 · Part 2 Core Electromagnetic spectrum [5]

Mark scheme

(a) **[2]**

- damage to surface (skin) cells, (which can lead to) skin cancer (1)
- damage to the eyes / eye conditions (1)

ALLOW sunburn / damage to the skin for the first mark

REJECT "skin burns" (that is the effect of infrared) as the only answer

(b) **[3]**

- the statement is incorrect / some lower-frequency radiation can also harm people (1)
- microwaves: internal heating of body cells (1)
- infrared: skin burns (1)
- the second and third marks need both the named region and its correct effect

Worked solution

Too much ultraviolet damages surface cells of the skin, which can lead to skin cancer, and it can damage the eyes, leading to eye conditions.

The student is wrong. Microwaves and infrared both have lower frequencies than visible light, and both can harm people with excessive exposure. Microwaves can cause internal heating of body cells, and infrared can cause skin burns. The higher-frequency regions (ultraviolet, X-rays, gamma rays) are harmful too, but they are not the only ones.

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25 · Part 2 Core Electromagnetic spectrum [5]

Mark scheme

(a) **[1]**

- microwaves (1)

(b) **[1]**

- geostationary (satellite) (1)

(c) **[2]**

- any two from: (1) each
  - mobile phones
  - microwave ovens

  ALLOW wireless internet

REJECT satellite television / satellite phones

(d) **[1]**

- any one from: radio / television transmissions; astronomy; radio frequency identification (RFID) (1)

Worked solution

Communication with artificial satellites is mainly by microwaves. Some satellite phones use low orbit satellites; direct broadcast satellite television uses geostationary satellites, which stay above the same point on the Earth. Microwaves are also used by mobile phones and in microwave ovens. Radio waves are used for radio and television transmissions, in astronomy and for RFID tags.

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26 · Part 2 Extended Electromagnetic spectrum [6]

Mark scheme

(a) **[1]**

- 3.0×10^8 m / s (1)

ALLOW 3×10^8 m / s

(b) **[3]**

- 2.5 GHz = 2.5×10^9 Hz (1)
- $\lambda = v / f$ or $\lambda = (3.0 \times 10^8) / (2.5 \times 10^9)$ (1)
- 0.12 m (1)

ALLOW 12 cm

ECF from the value of speed given in (a)

- if the frequency is not converted (e.g. 2.5 or 2.5×10^6 used), award the second mark only for correct use of $\lambda = v / f$

(c) **[2]**

- $f = v / \lambda$ or $f = (3.0 \times 10^8) / 3.2$ (1)
- 9.4×10^7 Hz (1)

ALLOW 94 MHz; 9.375×10^7 Hz

ECF from the value of speed given in (a)

Worked solution

All electromagnetic waves, including microwaves, travel at 3.0×10^8 m / s in a vacuum, and at approximately the same speed in air.

(b) Convert the frequency first: 2.5 GHz = 2.5×10^9 Hz. Rearranging $v = f\lambda$ gives $\lambda = v / f = (3.0 \times 10^8) / (2.5 \times 10^9) = 0.12$ m. Check: 0.12 m × 2.5×10^9 Hz = 3.0×10^8 m / s.

(c) $f = v / \lambda = (3.0 \times 10^8) / 3.2 = 9.375 \times 10^7$ Hz, which is 9.4×10^7 Hz to two significant figures (about 94 MHz). Check: 9.4×10^7 Hz × 3.2 m ≈ 3.0×10^8 m / s.

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27 · Part 2 Extended Electromagnetic spectrum [6]

Mark scheme

(a) **[3]**

- microwaves (1)
- (microwaves) can penetrate (some) walls (1)
- (microwaves) only require a short aerial (for transmission and reception) (1)

(b) **[1]**

- radio waves pass through walls but the signal is weakened as it does so (1)

ALLOW the walls absorb some of the signal

(c) **[2]**

- visible light and / or infrared (1)
- any one from: (1)
  - glass is transparent to visible light and some infrared
  - visible light and (short wavelength) infrared can carry high rates of data

Worked solution

Wireless internet, like mobile phones, uses microwaves. Microwaves can penetrate some walls, so the signal reaches other rooms, and they need only a short aerial, so the router and devices can be small.

Bluetooth uses radio waves. Radio waves pass through walls, but each wall weakens the signal, so the sound becomes less clear in another room.

Optical fibres carry visible light or infrared. Glass is transparent to visible light and some infrared, and these radiations can carry data at high rates, which is why fibres are used for high-speed broadband and cable television.

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28 · Part 2 Extended Electromagnetic spectrum [5]

Mark scheme

(a) **[2]**

- an analogue signal varies continuously / can have any value (within a range) (1)
- a digital signal can have only a limited number of values / only two values (e.g. on and off, 1 and 0) (1)

(b) **[2]**

- X: analogue (1)
- Y: digital (1)

(c) **[1]**

- not correct: sound can be transmitted as either a digital or an analogue signal (1)
- the mark needs both "not correct" and the reason

Worked solution

An analogue signal changes smoothly and can take any value in its range. A digital signal jumps between a small number of fixed values, usually just two, such as 0 and 1 (here 0 V and 5 V).

Signal X varies continuously and can take any value between −0.2 V and +0.2 V, so it is analogue. Signal Y only ever has the values 0 V or 5 V, so it is digital.

The student is wrong: a sound can be transmitted as either an analogue signal or a digital signal. Music on a digital radio or a phone call over the internet is sound sent digitally.

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29 · Part 2 Extended Electromagnetic spectrum [6]

Mark scheme

(a) **[2]**

- increased rate of transmission of data (1)
- increased range (1)

ALLOW can be sent further for increased range

ALLOW the signal can be regenerated accurately / noise can be removed, for the increased range mark

(b) **[3]**

- noise changes the shape / values of the signal (as it travels) (1)
- a digital signal has only two values (high and low / 1 and 0), so the regenerator can tell which value each part should be and remove the noise (1)
- an analogue signal can take any value, so the noise cannot be told apart from the signal and is amplified / kept with it (1)

(c) **[1]**

- the signal can be regenerated many times (along the cable) without noise building up, so it is still accurate after a long distance (1)

Worked solution

Digital signalling allows a higher rate of data transmission and a greater range.

As any signal travels along the cable it gets weaker and noise is added. A digital signal only has two allowed values, so at each regeneration point the equipment just decides whether each part is high or low and sends out a clean new copy; the noise is removed. An analogue signal can have any value, so the equipment cannot tell which changes are the real signal and which are noise. When it is amplified, the noise is amplified too.

Because a digital signal can be cleaned up accurately again and again, noise never builds up, so the signal can be sent over a much greater distance and still be accurate.

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30 · Part 2 Core Electromagnetic spectrum [6]

Mark scheme

(a) **[1]**

- gamma rays, visible light, radio waves (1)

(b) **[2]**

- they all arrive at the same time (1)
- all electromagnetic waves travel at the same (high) speed in a vacuum (and they set off at the same instant and travel the same distance) (1)
- the explanation mark can only be scored if the prediction is "at the same time"

(c) **[2]**

- any two from: (1) each
  - sterilising food
  - sterilising medical equipment
  - detection of cancer
  - treatment of cancer

> **ALLOW** "detection and treatment of cancer" for two marks; "treating cancer" / "radiotherapy" alone for one mark

(d) **[1]**

- any one from: security marking; detecting fake bank notes; sterilising water (1)

Worked solution

Gamma rays have the shortest wavelength and radio waves the longest, so in order of increasing wavelength the order is gamma rays, visible light, radio waves.

All electromagnetic waves travel at the same high speed in a vacuum. The three radiations left the star at the same instant and travelled the same distance at the same speed, so they reach the satellite at the same time.

On Earth, gamma rays are used to sterilise food and medical equipment, and to detect and treat cancer. Ultraviolet is used for security marking, for detecting fake bank notes and for sterilising water.

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1 · Part 3 Core Sound [1]

Answer: B

Explanation

Sound is produced by a vibrating source: the vibrating string makes the neighbouring air particles vibrate, and the vibration is passed on through the air. A is wrong because sound is not made of particles given out by the source. C is wrong because sound is not a transfer of warm air. D is wrong because the air vibrates backwards and forwards; it does not flow steadily away.

2 · Part 3 Core Sound [1]

Answer: D

Explanation

In a longitudinal wave the vibrations are parallel to the direction the wave travels. B describes a transverse wave. C is wrong because waves transfer energy, not matter: the particles only vibrate about fixed positions. A confuses the word "longitudinal" with distance.

3 · Part 3 Core Sound [1]

Answer: A

Explanation

Humans can hear roughly 20 Hz to 20 000 Hz, and 12 000 Hz lies inside this range. 8 Hz is below the range. 35 kHz (35 000 Hz) and 150 000 Hz are above it, so they are ultrasound.

4 · Part 3 Core Sound [1]

Answer: C

Explanation

Sound needs a medium (solid, liquid or gas) to travel through, and space is a vacuum. A is wrong because the frequency is not the problem. B is wrong because sound does not travel at all through a vacuum. D is wrong because the sound never reaches the helmet.

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5 · Part 3 Core Sound [1]

Answer: A

Explanation

The speed of sound in air is approximately 330–350 m / s, so 340 m / s is correct. B and C are ten times too small and ten times too large. D is the speed of light (electromagnetic waves), which is much faster than sound.

6 · Part 3 Core Sound [1]

Answer: B

Explanation

A louder sound has a larger amplitude, and a higher-pitched sound has a higher frequency. A has the frequency change the wrong way round. C has the amplitude change the wrong way round. D keeps the amplitude the same, which would give the same loudness.

7 · Part 3 Core Sound [1]

Answer: D

Explanation

With the settings unchanged, more complete waves on the screen means a higher frequency, so a higher pitch: B or D. A smaller peak height means a smaller amplitude, so a quieter sound: C or D. Only D has both. A is the loudest, lowest note; B is loud and high; C is quiet but low.

8 · Part 3 Core Sound [1]

Answer: C

Worked solution

total distance = v × t = 340 m / s × 0.50 s = 170 m. The sound goes to the rock face and back, so distance to the rock face = 170 m / 2 = 85 m. Check: 2 × 85 m = 170 m, and 170 m / 340 m / s = 0.50 s.

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9 · Part 3 Core Sound [1]

Answer: B

Explanation

An echo is the reflection of sound waves from a surface. A is wrong because refraction is a change of direction caused by a change of speed when entering a different medium. C is wrong because reflection does not change the frequency. D is wrong because the echo travels through the same air at the same speed.

10 · Part 3 Core Sound [1]

Answer: A

Explanation

Ultrasound is defined as sound with a frequency higher than 20 kHz (20 000 Hz), above the upper limit of human hearing. B describes sound below the range of hearing. C includes almost all audible sound. D is wrong because ultrasound is a sound wave, not an electromagnetic wave.

11 · Part 3 Core Sound [1]

Answer: D

Worked solution

v = s / t = 660 m / 2.0 s = 330 m / s. Check: 330 m / s × 2.0 s = 660 m. This lies inside the expected range of 330–350 m / s.

12 · Part 3 Extended Sound [1]

Answer: C

Explanation

A rarefaction is a region of a longitudinal wave where the particles are spread further apart than normal, so the pressure is lower. A describes a compression. B is wrong because particles vibrate everywhere along the wave. D confuses rarefaction with reflection.

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13 · Part 3 Extended Sound [1]

Answer: B

Explanation

In general sound travels fastest in solids, slower in liquids and slowest in gases. So the slowest speed (Y, 340 m / s) is air, the middle speed (X, 1480 m / s) is water and the fastest (Z, 5100 m / s) is steel. A and C give air or steel the middle speed. D gives steel the slowest speed.

14 · Part 3 Extended Sound [1]

Answer: A

Worked solution

Extra time for the seabed echo = 0.40 s − 0.24 s = 0.16 s. This extra time covers the gap between fish and seabed twice (down and back), so the gap = 1500 m / s × 0.16 s / 2 = 120 m. Check with the two depths: 300 m − 180 m = 120 m.

15 · Part 3 Extended Sound [1]

Answer: C

Explanation

Non-destructive testing means checking the inside of an object without damaging it. Ultrasound pulses sent into the rail are reflected by cracks, so hidden cracks can be found. A is a use of gamma radiation. B is a use of X-rays. D is a use of infrared radiation.

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16 · Part 3 Core Sound [5]

Mark scheme

(a) **[2]**

- the cone vibrates / moves backwards and forwards and pushes on the air particles next to it (1)
- the air particles vibrate and pass the vibration on to neighbouring particles (so the vibration travels through the air) (1)

(b) **[2]**

- longitudinal (1)
- particles vibrate / move backwards and forwards parallel to the direction the wave travels (1)

REJECT particles travel with the wave from the loudspeaker to the listener

(c) **[1]**

- yes, because 50 Hz is within the range 20 Hz to 20 000 Hz (1)

ALLOW yes, it is above the lowest frequency humans can hear (20 Hz)

Worked solution

(a) Sound is made by vibrating objects. As the cone moves forwards it pushes the air particles in front of it together; as it moves back they spread out. The air particles vibrate and pass this vibration from particle to particle, so a sound wave travels outwards.

(b) Sound is a longitudinal wave. Each air particle vibrates backwards and forwards about a fixed position, in the same direction as the wave is travelling. The particles do not travel to the listener; only the energy does.

(c) Humans with normal hearing can hear frequencies from about 20 Hz to 20 000 Hz. 50 Hz is inside this range, so it can be heard (as a low-pitched hum).

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17 · Part 3 Core Sound [5]

Mark scheme

(a) **[1]**

- the sound gets quieter (until it can hardly be heard / cannot be heard) (1)

(b) **[2]**

- sound needs a medium to travel through (1)
- as air is removed there are fewer particles to pass on the vibrations / a vacuum cannot transmit sound (1)

(c) **[1]**

- light can travel through a vacuum / light does not need a medium (1)

ALLOW light is an electromagnetic wave and can travel through a vacuum

(d) **[1]**

- some air is still left in the container / it is not a perfect vacuum (1)

ALLOW sound travels through the thread and the glass (solids) to the outside

Worked solution

(a), (b) Sound is a vibration passed from particle to particle, so it needs a medium. As the pump removes air, there are fewer air particles to carry the vibrations from the phone to the glass, so the sound gets quieter and quieter.

(c) The screen can still be seen, so light reaches the student through the vacuum. Light is an electromagnetic wave, and electromagnetic waves do not need a medium.

(d) A pump cannot remove every air particle, so a little sound can still pass through the remaining air. Also, the phone hangs from a thread, so some vibration can travel through the thread and the glass, which are solids.

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18 · Part 3 Core Sound [7]

Mark scheme

(a) **[4]**

- one student bangs the blocks together; the other stands a large distance away (e.g. 100 m or more) (1)
- measure the distance between the students with the tape measure (1)
- start the stopwatch on seeing the blocks hit and stop it on hearing the sound (1)
- speed = distance / time (1)
- repeat the timing several times and find the mean time (1)
- max 4

(b) **[3]**

- mean time = (0.52 + 0.55 + 0.53) / 3 = 0.533 s (1)
- speed = distance / time = 180 / 0.533 (1)
- 340 m / s (1)

ALLOW 337.5 m / s, 338 m / s

ECF from an incorrect mean time

Worked solution

(a) Stand the students far apart in an open space and measure the distance between them with the tape measure. One student bangs the blocks together. The other starts the stopwatch when he or she sees the blocks hit (light arrives almost instantly) and stops it when the sound is heard. Repeat several times and take the mean time. Then speed of sound = distance / mean time. A large distance makes the time longer, so the reaction time of the person timing matters less.

(b) Mean time = (0.52 s + 0.55 s + 0.53 s) / 3 = 1.60 s / 3 = 0.533 s. Speed = 180 m / 0.533 s = 337.5 m / s, which is 340 m / s to 2 significant figures. Check: 340 m / s × 0.533 s ≈ 181 m, close to 180 m.

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19 · Part 3 Core Sound [6]

Mark scheme

(a) **[1]**

- the reflection of sound (waves) (from a surface) (1)

(b) **[1]**

- 8.0 / 20 = 0.40 s (1)

(c) **[3]**

- distance travelled by the sound = 2 × 66 = 132 m (1)
- speed = distance / time = 132 / 0.40 (1)
- 330 m / s (1)

ECF from (b)

(d) **[1]**

- reduces the effect of the friend's reaction time (on the result) / the time is long, so the timing error is small compared with it (1)

ALLOW one interval is too short to time accurately with a stopwatch

Worked solution

(a) An echo is sound that has been reflected from a surface, here the school wall.

(b) From clap zero to clap 20 there are 20 equal intervals, so one interval = 8.0 s / 20 = 0.40 s.

(c) In one interval the sound from a clap travels to the wall and back, a distance of 2 × 66 m = 132 m. Speed = 132 m / 0.40 s = 330 m / s. Check: 330 m / s × 0.40 s = 132 m.

(d) Starting and stopping a stopwatch involves a reaction time of a few tenths of a second. That would be a huge error compared with 0.40 s, but it is small compared with 8.0 s.

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20 · Part 3 Core Sound [5]

Mark scheme

(a) **[2]**

- the peaks are higher (than 2.0 cm) / the trace has a larger amplitude (1)
- the same number of waves (4) across the screen / the same frequency (1)

(b) **[2]**

- R has a higher pitch (than P) (1)
- R has the same loudness as P (1)

(c) **[1]**

- frequency (1)

Worked solution

(a) Loudness depends on amplitude, so a louder sound gives taller peaks on the trace. Pitch depends on frequency; Q has the same pitch, so it has the same frequency and the trace still shows 4 complete waves.

(b) R shows 8 waves in the same time instead of 4, so its frequency is twice as high and its pitch is higher. The peaks are the same height, so the amplitude, and therefore the loudness, is the same as P.

(c) The pitch of a sound depends on its frequency: the higher the frequency, the higher the pitch.

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21 · Part 3 Core Sound [5]

Mark scheme

(a) **[1]**

- 20 Hz to 20 000 Hz (1)

(b) **[2]**

- 15 Hz (1)
- 30 000 Hz (1)

(c) **[2]**

- ultrasound (1)
- sound with a frequency higher than 20 kHz / 20 000 Hz (1)

Worked solution

(a) A young person with normal hearing can hear sounds from about 20 Hz to 20 000 Hz.

(b) 15 Hz is below 20 Hz, so it is too low to hear. 30 000 Hz is above 20 000 Hz, so it is too high to hear. 250 Hz, 4000 Hz and 18 000 Hz are all inside the range.

(c) Sound above the range of human hearing is called ultrasound. It is defined as sound with a frequency higher than 20 kHz. 30 000 Hz is 30 kHz, so it is ultrasound.

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22 · Part 3 Core Sound [6]

Mark scheme

(a) **[1]**

- the sound (wave) is reflected from the iceberg (1)

(b) **[3]**

- total distance travelled = 340 × 3.0 = 1020 m (1)
- the sound travels there and back, so divide by 2 (1)
- 510 m (1)

ALLOW 1 mark only for 1020 m as the final answer

(c) **[2]**

- the time gets shorter (1)
- the sound has a smaller distance to travel (at the same speed) (1)

Worked solution

(a) The sound waves from the horn travel to the iceberg and are reflected back to the ship. The reflected sound is the echo.

(b) In 3.0 s the sound travels distance = speed × time = 340 m / s × 3.0 s = 1020 m. This is the distance to the iceberg and back again, so the iceberg is 1020 m / 2 = 510 m away. Check: 2 × 510 m = 1020 m, and 1020 m / 340 m / s = 3.0 s.

(c) As the ship gets closer, the sound has a shorter distance to travel to the iceberg and back. The speed of sound in the air does not change, so the echo returns sooner.

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23 · Part 3 Core Sound [6]

Mark scheme

(a) **[1]**

- light travels (very much) faster than sound (1)

(b) **[2]**

- distance = speed × time = 340 × 4.5 (1)
- 1500 m (1)

> **ALLOW** 1530 m

(c) **[3]**

- new distance = 340 × 3.0 = 1020 m (1)
- distance moved by the storm = 1530 − 1020 = 510 m (1)
- average speed = 510 / 600 = 0.85 m / s (1)

> **ALLOW** 0.80 m / s from using 1500 m

> **ALLOW** 0.83 m / s from using 1500 m and 1000 m

> **ECF** from (b)

Worked solution

(a) The light from the lightning reaches the student almost instantly, because light travels much faster than sound. The sound of the thunder takes a few seconds to arrive.

(b) The time delay is almost all the time taken by the sound. Distance = 340 m / s × 4.5 s = 1530 m, which is 1500 m to 2 significant figures.

(c) For the second flash, distance = 340 m / s × 3.0 s = 1020 m. The storm has moved 1530 m − 1020 m = 510 m closer. Ten minutes = 10 × 60 = 600 s. Average speed = total distance / total time = 510 m / 600 s = 0.85 m / s. Check: 0.85 m / s × 600 s = 510 m.

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24 · Part 3 Core Sound [6]

Mark scheme

(a) **[3]**

- the prongs of the tuning fork vibrate (1)
- the prongs make the air particles next to them vibrate (1)
- the vibrations are passed from particle to particle through the air to the ear / a longitudinal wave travels through the air (1)

ALLOW the air particles do not travel to the ear, only the energy does, for the third mark

(b) **[2]**

- in the sound wave the particles vibrate parallel to the direction the wave travels (1)
- in the water wave the particles vibrate at right angles (perpendicular) to the direction the wave travels (1)

(c) **[1]**

- the table top is also made to vibrate, and its large surface makes more air vibrate / gives a larger amplitude (1)

Worked solution

(a) Sound is produced by vibrating sources. The prongs of the tuning fork vibrate quickly. Each time a prong moves outwards it pushes the air particles next to it, and these particles push on the next ones, so the vibration passes through the air as a longitudinal wave until it makes the eardrum vibrate. The air particles only vibrate about their positions; energy is transferred, not air.

(b) Both waves are vibrations that transfer energy. In sound (longitudinal) the particles move backwards and forwards along the direction the wave travels. In the water wave (transverse) the particles move up and down, at right angles to the direction the wave travels.

(c) The vibrating handle makes the table top vibrate too. The table has a much larger surface than the prongs, so it pushes on much more air and the sound wave has a larger amplitude, so the note is louder.

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25 · Part 3 Core Sound [6]

Mark scheme

(a) **[1]**

- the amplitude (of the sound wave) is larger (1)

IGNORE references to the frequency staying the same

(b) **[3]**

- λ = v / f (1)
- 330 / 440 (1)
- 0.75 m (1)

(c) **[1]**

- the pitch is higher (1)

(d) **[1]**

- 0.50 m (1)

ALLOW 0.5 m

Worked solution

(a) A louder sound has a larger amplitude. The pitch has not changed, so the frequency is the same.

(b) Wave speed v = fλ, so λ = v / f = 330 m / s / 440 Hz = 0.75 m. Check: 440 Hz × 0.75 m = 330 m / s.

(c) A higher frequency gives a higher pitch, so the note sounds higher.

(d) The speed of sound in the same air is still 330 m / s, so λ = 330 m / s / 660 Hz = 0.50 m. The frequency has been multiplied by 1.5, so the wavelength is divided by 1.5: 0.75 m / 1.5 = 0.50 m.

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26 · Part 3 Core Sound [7]

Mark scheme

(a) **[1]**

- the time increases (in proportion) / doubling the distance (roughly) doubles the time (1)

(b) **[3]**

- distance travelled by the sound = 2 × 80 = 160 m (1)
- speed = distance / time = 160 / 0.47 (1)
- 340 m / s (1)

ALLOW 340.4 m / s

(c) **[2]**

- time = distance / speed = (2 × 100) / 340 (1)
- 0.59 s (1)

ALLOW 0.58 s to 0.60 s

ECF from (b)

(d) **[1]**

- the times are too short to measure accurately with a stopwatch / human reaction time is large compared with the times measured (1)

Worked solution

(a) Each time the distance goes up by 20 m, the echo time goes up by about 0.12 s. The time is (roughly) proportional to the distance.

(b) The sound travels to the wall and back, so it travels 2 × 80 m = 160 m in 0.47 s. Speed = 160 m / 0.47 s = 340 m / s (to 2 significant figures).

(c) At 100 m the sound travels 2 × 100 m = 200 m. Time = 200 m / 340 m / s = 0.588 s, which is 0.59 s. Check: the times rise by about 0.12 s per 20 m, and 0.47 s + 0.12 s = 0.59 s.

(d) The echo times are less than half a second. A person's reaction time when using a stopwatch is a few tenths of a second, which would cause a very large error. An electronic timer triggered by the microphone has no reaction time.

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27 · Part 3 Extended Sound [6]

Mark scheme

(a) **[2]**

- (i) a region where the air particles are closer together than normal / a region of higher pressure (1)
- (ii) a region where the air particles are further apart than normal / a region of lower pressure (1)

(b) **[2]**

- as the cone moves forwards it pushes the air particles together, producing a compression (1)
- as the cone moves backwards the air particles spread out, producing a rarefaction (1)

(c) **[2]**

- $f = v / \lambda = 340 / 0.68$ (1)
- 500 Hz (1)

Worked solution

(a) Sound is a longitudinal wave. In a compression the particles are squashed closer together than normal, so the pressure is higher. In a rarefaction the particles are spread further apart than normal, so the pressure is lower.

(b) When the cone moves forwards, it pushes the air in front of it, squeezing the particles together into a compression. When it moves back, it leaves more space, so the particles spread out into a rarefaction. Each vibration of the cone sends out one compression and one rarefaction.

(c) The distance between the centres of neighbouring compressions is one wavelength, so $\lambda = 0.68$ m. From $v = f\lambda$, $f = v / \lambda = 340$ m / s / 0.68 m = 500 Hz. Check: 500 Hz × 0.68 m = 340 m / s.

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28 · Part 3 Extended Sound [6]

Mark scheme

(a) **[2]**

- the sound through the steel arrives first (1)
- sound travels faster in steel (solids) than in air (gases) (1)

(b) **[3]**

- time through air = 102 / 340 = 0.30 s (1)
- time through steel = 102 / 5100 = 0.020 s (1)
- time interval = 0.30 − 0.020 = 0.28 s (1)

ECF from either time

(c) **[1]**

- sound is (in general) fastest in solids, slower in liquids and slowest in gases (1)

Worked solution

(a) Both sounds travel the same 102 m, but sound travels much faster in a solid such as steel than in a gas such as air, so the sound through the handrail arrives first.

(b) Time = distance / speed. Through the air: 102 m / 340 m / s = 0.30 s. Through the steel: 102 m / 5100 m / s = 0.020 s. The time between the two sounds is 0.30 s − 0.020 s = 0.28 s. Check: 340 m / s × 0.30 s = 102 m and 5100 m / s × 0.020 s = 102 m.

(c) In general, sound travels fastest in solids, more slowly in liquids and most slowly in gases.

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29 · Part 3 Extended Sound [7]

Mark scheme

(a) **[2]**

- the pulse travels down and is reflected from the seabed (1)
- the time between sending the pulse and detecting the echo is measured, and depth = (speed × time) / 2 (1)

ALLOW the time is used with the speed of ultrasound in water to calculate the distance

(b) **[3]**

- distance travelled = 1500 × 0.064 = 96 m (1)
- divide by 2 for there and back (1)
- 48 m (1)

ALLOW 1 mark only for 96 m as the final answer

(c) **[2]**

- the time gets longer (1)
- the pulse travels the same distance (down and back) at a lower speed (1)

Worked solution

(a) The ship sends a short pulse of ultrasound down through the water. The pulse reflects from the seabed and the echo returns to a detector on the ship. The system measures the time between sending the pulse and receiving the echo. Since the pulse travels down and back, depth = speed × time / 2.

(b) Distance travelled = 1500 m / s × 0.064 s = 96 m. The pulse went down and back, so the depth = 96 m / 2 = 48 m. Check: 2 × 48 m / 1500 m / s = 0.064 s.

(c) The distance to the seabed and back is the same, but the ultrasound now moves more slowly, so time = distance / speed is larger and the echo arrives later.

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30 · Part 3 Extended Sound [7]

Mark scheme

(a) **[2]**

- a pulse of ultrasound is sent into the block and is reflected by the crack (1)
- an echo is detected sooner than the echo from the bottom surface (because the crack is closer) (1)

ALLOW an extra echo is detected for the second mark

(b) **[2]**

- time = distance / speed = (2 × 0.32) / 6400 (1)
- 1.0×10^{-4} s (1)

ALLOW 0.00010 s, 100 μs

(c) **[2]**

- distance travelled = $6400 \times 25 \times 10^{-6} = 0.16$ m (1)
- depth = 0.16 / 2 = 0.080 m (1)

ALLOW 0.08 m, 8.0 cm

(d) **[1]**

- X-rays are ionising and can damage cells / the fetus, but ultrasound does not (1)

ALLOW X-rays can cause mutations or cancer

ALLOW ultrasound gives images of soft tissue

Worked solution

(a) The probe sends a short pulse of ultrasound into the block. In a good block the only echo comes from the bottom surface. If there is a crack, some ultrasound is reflected from it, so an extra echo arrives earlier than the bottom-surface echo. The block is not damaged, so this is non-destructive testing.

(b) The pulse travels down to the bottom and back: 2 × 0.32 m = 0.64 m. Time = 0.64 m / 6400 m / s = 1.0×10^{-4} s (100 μs).

(c) 25 μs = 25×10^{-6} s. Distance travelled = 6400 m / s × 25×10^{-6} s = 0.16 m. This is down to the crack and back, so the crack is 0.16 m / 2 = 0.080 m below the top surface. Check: the crack echo arrives at 25 μs out of 100 μs, so the crack is a quarter of the way down: 0.32 m / 4 = 0.080 m.

(d) X-rays are ionising radiation and could harm the cells of the developing baby. Ultrasound is a sound wave, is not ionising, and gives clear images of soft tissue.

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